Basics of matrices

Basic skills

Q1. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ 2 rows, 2 cols
1's on diag

Q2. $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ 4 rows, 3 cols

Q3. a) $\begin{pmatrix} 3 & 4 \\ 0 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 5 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 9 \\ 2 & 3 \end{pmatrix}$

b) $\begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 1 & 0 \end{pmatrix}$

c) $3\begin{pmatrix} 2 & -1 \\ 4 & 0 \end{pmatrix} = \begin{pmatrix} 6 & -3 \\ 12 & 0 \end{pmatrix}$

Q4. To add or subtract matrices they must be the same order as +/- each element space

Competency skills

Q5. Square matrix

Q6. $A = B$ if : • same order

• All elements equal

$a_{ij} = b_{ij}$ for all i, j

Q7. a)

$$2\begin{pmatrix} 5 & 2 \\ -1 & 4 \end{pmatrix} - 4\begin{pmatrix} -2 & 10 \\ 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 10 & 4 \\ -2 & 8 \end{pmatrix} - \begin{pmatrix} -8 & 40 \\ 4 & 0 \end{pmatrix} = \begin{pmatrix} 18 & -36 \\ -6 & 8 \end{pmatrix}$$

b)

$$\begin{pmatrix} -1 & 2 & -5 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{pmatrix} + 3\begin{pmatrix} 0 & 3 & 5 \\ -2 & 1 & -2 \\ -4 & -3 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 2 & -5 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{pmatrix} + \begin{pmatrix} 0 & 9 & 15 \\ -6 & 3 & -6 \\ -12 & -9 & 18 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 11 & 10 \\ -4 & 2 & -6 \\ -12 & -8 & 22 \end{pmatrix}$$

c) $\begin{pmatrix} 2 & 5 & 3 \\ -1 & 6 & 0 \end{pmatrix} - 2\begin{pmatrix} 5 & -4 & -1 \\ -2 & -1 & 6 \end{pmatrix}$

$= \begin{pmatrix} 2 & 5 & 3 \\ -1 & 6 & 0 \end{pmatrix} - \begin{pmatrix} 10 & -8 & -2 \\ -4 & -2 & 12 \end{pmatrix} = \begin{pmatrix} -8 & 13 & 5 \\ 3 & 8 & -12 \end{pmatrix}$

Q8. $\begin{pmatrix} x^2 & 5 \\ 2 & y \end{pmatrix} - 3\begin{pmatrix} x & -3 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2x & 14 \\ -1 & 4 \end{pmatrix}$

$\Rightarrow \quad x^2 - 3x = 2x, \quad y - 6 = 4$

so $\quad x^2 - 5x = 0, \quad y = 10$

$x(x-5) = 0$ so $x = 0, 5$

Q9. $\begin{pmatrix} y & 2x \\ 2 & 0 \end{pmatrix} - \begin{pmatrix} 1 & y \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} x & 1 \\ 0 & -2 \end{pmatrix}$

$\Rightarrow \quad y - 1 = x, \quad 2x - y = 1$ so $y = x + 1$

$\Rightarrow \quad 2x - (x+1) = 1 \quad \Rightarrow \quad x = 2$ so $y = 3$

Advanced skills

Q10. a) $\begin{pmatrix} \overset{a_{11}}{-1} & 0 & -5 \\ 2 & \overset{a_{22}}{3} & 2 \\ -4 & -3 & \overset{a_{33}}{6} \end{pmatrix}$

So trace $= -1 + 3 + 6$
$= 8$

b) As $A+B = \begin{pmatrix} a_{11}+b_{11} & & \\ & \ddots & \\ & & a_{nn}+b_{nn} \end{pmatrix}$

$$tr(A+B) = \sum_{i=1}^{n} (a_{ii}+b_{ii}) = \sum_{i=1}^{n} a_{ii} + \sum_{i=1}^{n} b_{ii}$$
$$= tr(A) + tr(B)$$

In words as tracing adds then adding can be done in this order of A's elements then B's elements

c) $kA = \begin{pmatrix} ka_{11} & & \\ & \ddots & \\ & & ka_{nn} \end{pmatrix}$

$$tr(kA) = \sum_{i=1}^{n} ka_{ii} = k \sum_{i=1}^{n} a_{ii} = k\,tr(A)$$

multipling each then adding is same as adding then multiplying

Q11. As $a+b=b+a \Rightarrow A+B=B+A$

Q12. True as set order to have 1 col (or row) eg $\begin{pmatrix}1\\2\end{pmatrix}$ is matrix of 2×1